

Dirac equation 3

Let

$$\begin{aligned}\psi_1 &= \sqrt{E + mc^2} \begin{pmatrix} 1 \\ 0 \\ \frac{p_z c}{E + mc^2} \\ \frac{(p_x + ip_y)c}{E + mc^2} \end{pmatrix} e^{-i\xi/\hbar} & \psi_2 &= \sqrt{E + mc^2} \begin{pmatrix} 0 \\ 1 \\ \frac{(p_x - ip_y)c}{E + mc^2} \\ \frac{-p_z c}{E + mc^2} \end{pmatrix} e^{-i\xi/\hbar} \\ &\text{wavefunction for fermion spin up} & & \text{wavefunction for fermion spin down} \\ \psi_3 &= \sqrt{E + mc^2} \begin{pmatrix} \frac{p_z c}{E + mc^2} \\ \frac{(p_x + ip_y)c}{E + mc^2} \\ 1 \\ 0 \end{pmatrix} e^{i\xi/\hbar} & \psi_4 &= \sqrt{E + mc^2} \begin{pmatrix} \frac{(p_x - ip_y)c}{E + mc^2} \\ \frac{-p_z c}{E + mc^2} \\ 0 \\ 1 \end{pmatrix} e^{i\xi/\hbar} \\ &\text{wavefunction for antifermion spin up} & & \text{wavefunction for antifermion spin down}\end{aligned}$$

where

$$\xi = p_\mu x^\mu = Et - p_x x - p_y y - p_z z$$

and

$$E = \sqrt{p_x^2 c^2 + p_y^2 c^2 + p_z^2 c^2 + m^2 c^4}$$

Verify the wavefunctions solve the Dirac equation

$$i\hbar \left(\gamma^0 \frac{\partial}{\partial t} + c\gamma^1 \frac{\partial}{\partial x} + c\gamma^2 \frac{\partial}{\partial y} + c\gamma^3 \frac{\partial}{\partial z} \right) \psi = mc^2 \psi$$

and are normalized as

$$|\psi|^2 = 2E$$

Verify dimensions.

$$\begin{aligned}\frac{Et}{\hbar} &= \frac{[\text{J}] [\text{s}]}{[\text{J s}]} = [1] \\ \frac{p_x x}{\hbar} &= \frac{[\text{kg m s}^{-1}] [\text{m}]}{[\text{J s}]} = [1] \\ \hbar \frac{\partial}{\partial t} &= [\text{J s}] [\text{s}^{-1}] = [\text{J}] \\ \hbar c \frac{\partial}{\partial x} &= [\text{J s}] [\text{m s}^{-1}] [\text{m}^{-1}] = [\text{J}]\end{aligned}$$

Eigenmath script